Satellite Image Land-Use and Land-Cover Classification

CNN-based Classification on the EuroSAT Dataset

1. Project Objective

The goal of this project was to train a Convolutional Neural Network (CNN) to classify satellite images into one of ten land-use / land-cover categories, using the EuroSAT dataset. Each 64x64 RGB satellite image is classified as a whole into a single category (image-level classification, not pixel-level segmentation).

2. Dataset

- Dataset: EuroSAT RGB (Sentinel-2 derived satellite imagery)
- Total images: 27,000
- Classes (10): AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake
- Split used: 80% training (21,600 images) / 20% testing (5,400 images)
- Image size: 64 x 64 pixels, normalized RGB channels

3. Model Architecture

A custom CNN was built with three convolutional blocks (Conv2D + BatchNorm + ReLU + MaxPool), progressively extracting features from edges to complex land-cover patterns, followed by a fully connected classifier head with dropout (0.5) to reduce overfitting.

- Conv Block 1: 3 -> 32 channels
- Conv Block 2: 32 -> 64 channels
- Conv Block 3: 64 -> 128 channels
- Classifier: Flatten -> Linear(8192, 512) -> ReLU -> Dropout(0.5) -> Linear(512, 10)
- Loss function: CrossEntropyLoss | Optimizer: Adam (lr = 0.001) | Epochs: 10

4. Training Results

The model was trained for 10 epochs on a Kaggle GPU (T4 x2). Training loss decreased consistently from 1.18 to 0.33, with training accuracy improving from 59.75% to 89.27%, indicating stable and effective learning with no signs of divergence.

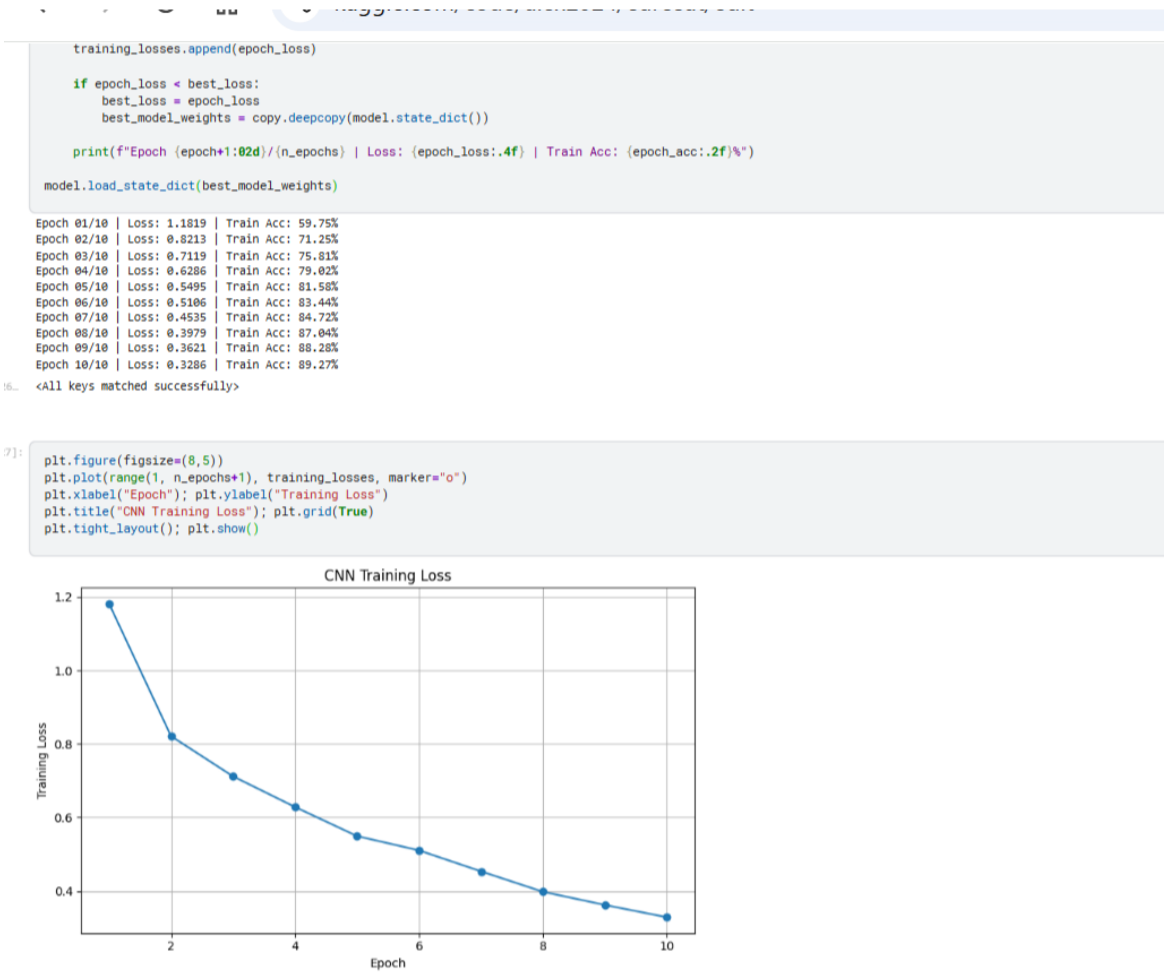

Figure 1: Per-epoch training log and training loss curve over 10 epochs.

5. Test Set Evaluation

On the held-out test set of 5,400 images, the trained model achieved an overall test accuracy of 88.19%. The table below shows per-class precision, recall, and F1-score.

Class	Precision	Recall	F1-score
AnnualCrop	0.93	0.90	0.91
Forest	0.75	0.99	0.85
HerbaceousVegetation	0.89	0.82	0.85
Highway	0.93	0.85	0.89
Industrial	0.97	0.89	0.93
Pasture	0.85	0.92	0.88
PermanentCrop	0.86	0.85	0.86
Residential	0.88	0.99	0.93
River	0.88	0.85	0.86
SeaLake	0.96	0.76	0.85

Overall macro average: Precision 0.89 | Recall 0.88 | F1-score 0.88, on 5,400 test images (Table generated from scikit-learn's classification_report).

```python
model.eval()
all_true, all_pred = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images, labels = images.to(device), labels.to(device)
        outputs = model(images)
        preds = outputs.argmax(1)
        all_true.extend(labels.cpu().numpy())
        all_pred.extend(preds.cpu().numpy())

test_acc = accuracy_score(all_true, all_pred)
print(f"Test Accuracy: {test_acc*100:.2f}%")
print(classification_report(all_true, all_pred, target_names=classes, zero_division=0))
```

```
Test Accuracy: 88.19%
                      precision    recall  f1-score   support

          AnnualCrop       0.93      0.90      0.91       631
              Forest       0.75      0.99      0.85       582
HerbaceousVegetation       0.89      0.82      0.85       612
             Highway       0.93      0.85      0.89       525
          Industrial       0.97      0.89      0.93       484
             Pasture       0.85      0.92      0.88       396
       PermanentCrop       0.86      0.85      0.86       506
         Residential       0.88      0.99      0.93       606
               River       0.88      0.85      0.86       497
             SeaLake       0.96      0.76      0.85       561

            accuracy                           0.88      5400
           macro avg       0.89      0.88      0.88      5400
        weighted avg       0.89      0.88      0.88      5400
```

Figure 2: Test accuracy and full classification report as generated in the notebook.

6. Confusion Matrix

The confusion matrix below shows actual vs predicted classes across the test set. The strong diagonal confirms most predictions are correct. Notable confusions include SeaLake being misclassified as Forest in 120 cases, and PermanentCrop/AnnualCrop being confused with HerbaceousVegetation — a known difficulty in the EuroSAT dataset due to visual similarity between vegetation types.

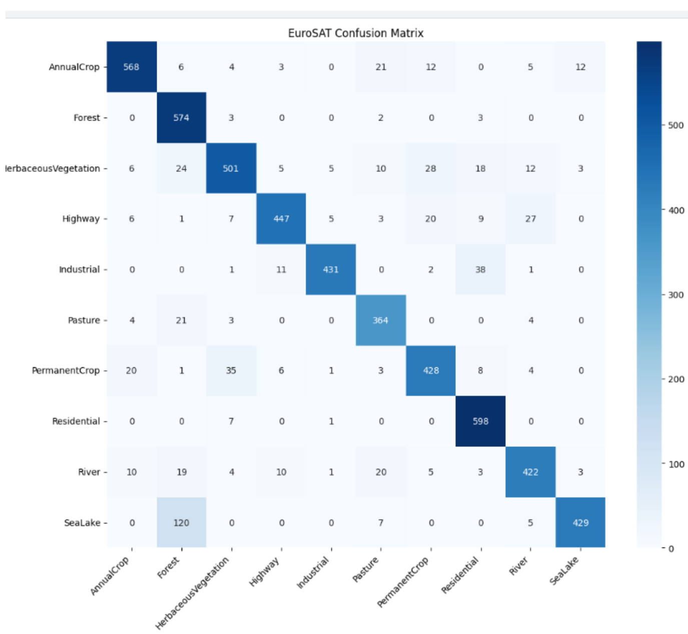

Figure 3: Confusion matrix across all 10 land-use classes on the test set.

7. Sample Predictions on New Images

To validate the saved model, sample images from each class folder were passed through the prediction pipeline. All four samples below were classified correctly with high confidence.

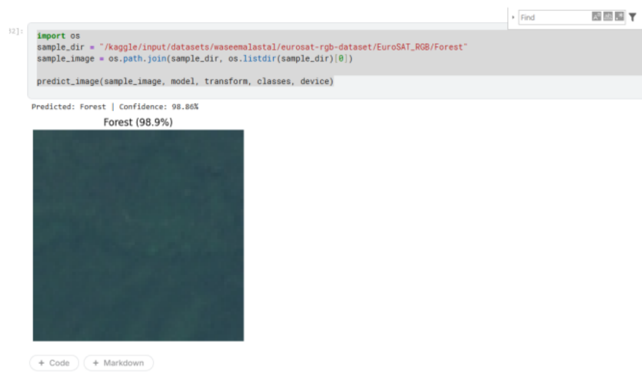

Figure 4a: Forest sample - Predicted 'Forest' with 98.86% confidence.

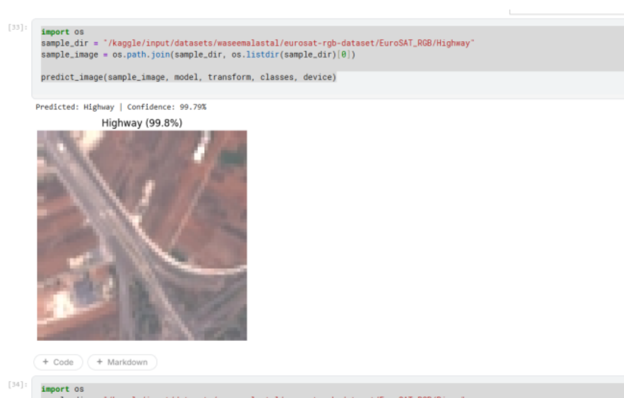

Figure 4b: Highway sample - Predicted 'Highway' with 99.79% confidence.

Figure 4c: River sample - Predicted 'River' with 63.88% confidence (lower confidence, reflecting visual overlap with Highway/AnnualCrop classes).

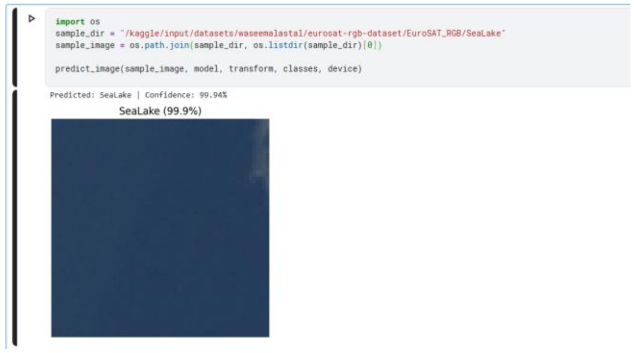

Figure 4d: SeaLake sample - Predicted 'SeaLake' with 99.94% confidence.

8. Summary

- Final test accuracy: 88.19% across 10 land-use classes
- Best-performing classes: Residential, Industrial, SeaLake (precision > 0.95)
- Weakest performing class: Forest (lower precision, 0.75) due to confusion with SeaLake
- Model saved as eurosat_cnn_model.pth for reuse without retraining

9. Limitations

This model performs whole-image classification only — it assigns a single label to an entire 64x64 image and does not identify multiple land-cover types within the same image at the pixel level. Pixel-level analysis would require a semantic segmentation model and a segmentation-labeled dataset.